

Program 1: Design and Implementation of a 4-to-2 Encoder Using VHDL

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity encoder is
    port(
        A :in STD_LOGIC_VECTOR(3 downto 0);
        B :out STD_LOGIC_VECTOR(1 downto 0)
    );
end encoder;

architecture Behavioral of encoder is
begin
    process (A)
    begin
        case A is
            when "0001" => B <= "00";
            when "0010" => B <= "01";
            when "0100" => B <= "10";
            when "1000" => B <= "11";
            when others => B <= "XX";
        end case;
    end process;
end Behavioral;
```

Program 2: Design and Implementation of a 2-to-4 Decoder Using VHDL

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity decoder is
    port(
        C :in STD_LOGIC_VECTOR(1 downto 0);
        D :out STD_LOGIC_VECTOR(3 downto 0)
    );
end decoder;

architecture Behavioral of decoder is
begin
    process (C)
    begin
        case C is
            when "00" => D <= "0001";
            when "01" => D <= "0010";
            when "10" => D <= "0100";
            when "11" => D <= "1000";
            when others => D <= "XXXX";
        end case;
    end process;
end Behavioral;
```

Figure 1: VHDL Code for 4-to-2 Encoder

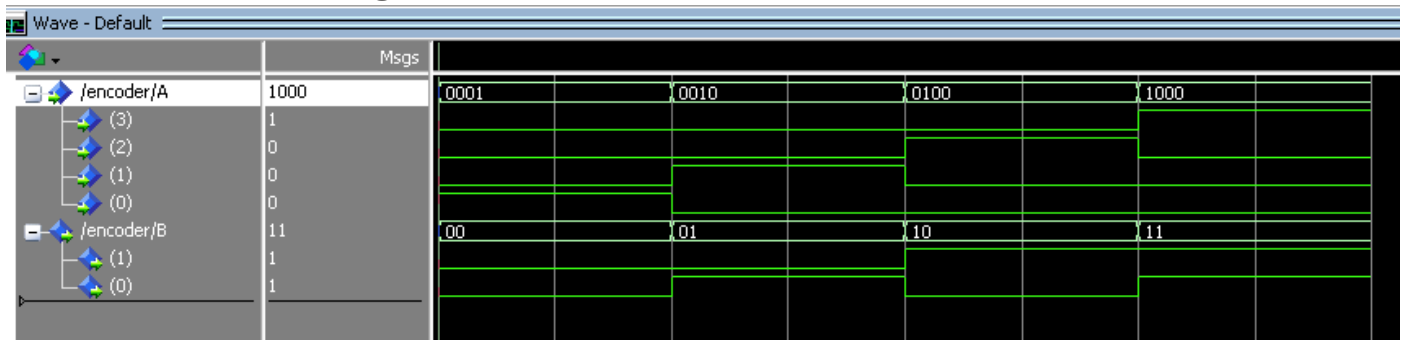


Figure 2: VHDL Code for 2-to-4 Decoder

